AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 06

TASOON Field Cultivator
Model: TS-700K

College of Engineering and Agro-Industrial Technology
University of the Philippines Los Baños,
College, Laguna, 4031 Philippines

Tel/Fax No. (049) 536-2527
e-mail: amtec@uplb.edu.ph
http://amtec.uplb.edu.ph

AMPC 18 x 24 TRAILING HARROW

Test Report No.	:	2013 - 29
Date of Test	:	March 14, 2013
Test Valid Until	:	March 14, 2018
Brand	:	AMPC
Model	:	18 x 24
Type	:	Offset gang, trailing type
Manufacturer	:	ACTION MULTI PURPOSE COOPERATIVE Turayong, Cauayan City Isabela
Test Requested by	:	Manufacturer
Machine Classification	:	Commercial

SUMMARY

The AMPC 18 x 24 Trailing Harrow was tested as a primary tillage implement using a sandy-loam type of soil at a farmer's field in Brgy. Murong, Cauayan City, Isabela. Hitched to a 64.2 kW MASSEY FERGUSSON 445 Four-Wheel Tractor, the Actual Field Capacity was 1.639 ha/h at a Travelling Speed of 9.46 kph with Field Efficiency of 89.9%. The average Depth and Width of cut were 135 mm and 2325 mm, respectively. Fuel Consumption of the tractor was 24.0 L/h.

PURPOSE AND SCOPE OF TEST

The implement was tested for verification of its field performance as a primary tillage implement for areas to be planted with corn. The field performance test was conducted at a farmer's field in Brgy. Murong, Cauayan City, Isabela on March 14, 2013. To verify and assess the implement specifications, work capacity, quality, field efficiency and adaptability of the implement, the test was carried out.

The field performance test was conducted by using a 64.2 kW MASSEY FERGUSSON 445 Four-Wheel Tractor. It was tested as a primary tillage implement under first pass operation. The items measured and observed were the following:

1. Depth and width of plowing
2. Actual traveling speed
3. Actual operating time
4. Fuel consumption
5. Evenness of plowed surface
6. Stability of the tractor during plowing operation

METHOD OF TEST

The machine was tested based on the "PAES 133:2004 Agricultural Machinery – Disc Harrow – Methods of Test".

OPERATOR'S MANUAL

No operator's manual or brochure was submitted with the implement.

DESCRIPTION OF THE IMPLEMENT

The AMPC 18 x 24 was trailing-type disc harrow with offset gang arrangement (**Figure 1**). It consisted of two gangs of discs arranged in a fixed-angle frame so that one gang runs behind the other. Each gang consisted of nine discs (**Figure 2**) which had a diameter of 610 mm. The discs were spaced 230 mm apart and each one was provided with a scraper (**Figure 3**) to clean and remove soil that may stick on the concave side of the disc blade.

The trailing harrow was provided with a hydraulic cylinder (**Figure 4**) which was connected to the hydraulic system of the tractor during operation. The hydraulic jack was used to lower the wheels during transport and to lift the wheels to serve as depth gauge of the harrow during operation.

A hitch point (**Figure 5**) was also provided to hitch the disc harrow to the drawbar of the tractor during operation.

The main frame (**Figure 6**) of the harrow consisted of a tubular cylinder (102 mm x 102 mm) with thickness of 5 mm.

Detailed specifications of the machine are shown in **Table 1**.

Figure 1. AMPC Trailing Harrow
Model: 18 x 24

Figure 2. A gang with nine discs

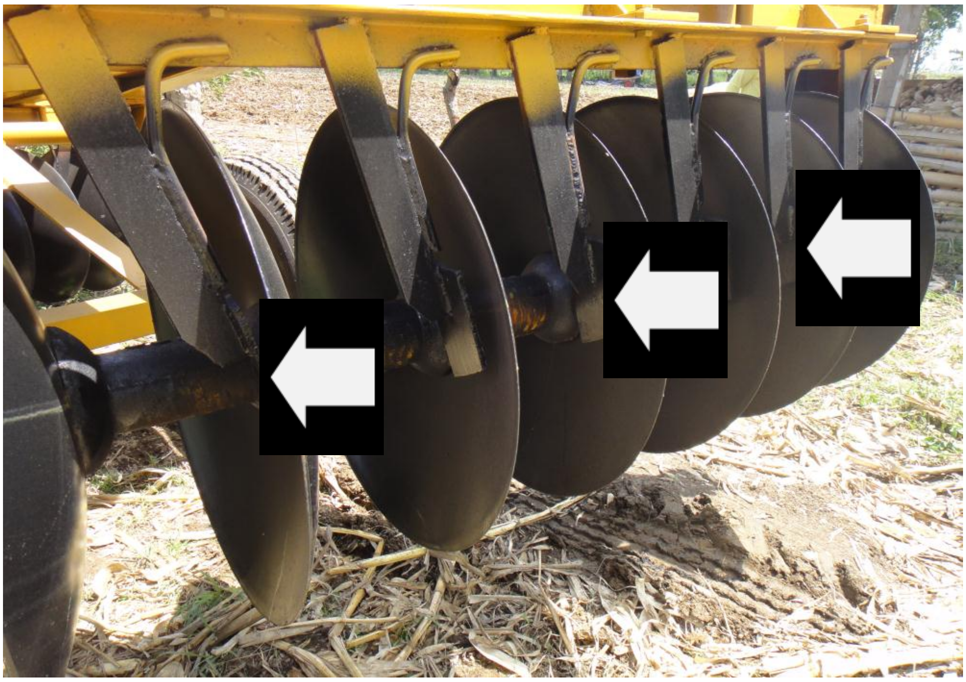

Figure 3. Scrapers

Figure 4. Hydraulic cylinder

Figure 5. Hitch-point

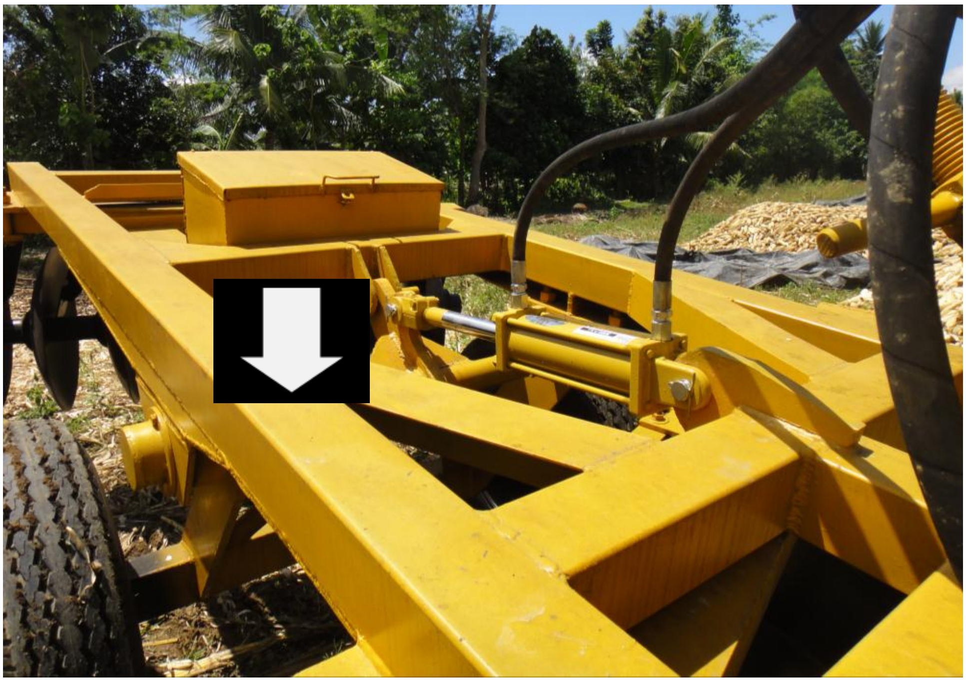

Figure 6. Main frame

IMPLEMENT SPECIFICATIONS

Table 1. Detailed specifications of the AMPC 18 x 24 trailing disc harrow

1.	Brand		:	AMPC
2.	Model		:	18 x 24
3.	Manufacturer		:	ACTION MULTI PURPOSE COOPERATIVE Turayong, Cauayan City
4.	Dimensions and weight			
	4.1	Overall length (mm)	:	4980
	4.2	Overall width (mm)	:	2400
	4.3	Overall height (mm)	:	1530
	4.4	Weight (kg)	:	1375
5.	Gang			
	5.1	Quantity	:	2
	5.2	Angle of front gang (deg)	:	16
	5.3	Angle of rear gang (deg)	:	16
	5.2	No. of discs/gang	:	9
	5.3	Disc spacing (mm)	:	230
	5.4	Gang axle bearing		
		5.4.1 No. of pcs./gang	:	6
		5.4.2 Type	:	Ball Bearing #6310

Table 1. Detailed specifications of the AMPC 18 x 24 trailing harrow (cont'n.)

6.	Disc			
	6.1	Brand	:	NIAUX
	6.2	Make	:	France
	6.3	Diameter (mm)	:	610
	6.4	Thickness (mm)	:	5.0
	6.5	Concavity (mm)	:	84
7.	Weight per disc (kg)		:	76.4
8.	Scraper			
	8.1	Dimensions, L x W x T (mm)	:	178 x 102 x 7
	8.2	Material	:	Mild Steel
9.	Spool			
	9.1	Dimensions, L x D (mm)	:	229 x 165
	9.2	Material	:	Cast Iron
10.	Frame			
	10.1	Make	:	Local
	10.2	Material	:	Tubular cylinder (102 mm x 102 mm x 5 mm)
11.	Tires			
	11.1	Quantity	:	2
	11.2	Size	:	800 x 190

RESULTS AND DISCUSSIONS

The AMPC 18 x 24 Trailing Disc Harrow was tested as a primary tillage implement (for areas to be planted with corn) in a sandy loam type of soil at a farmer's field in Barangay Murong, Cauayan City, Isabela.

Results of the performance tests are shown in **Table 2**.

Field Capacity

Hitched to a 64.2 kW MASSEY FERGUSSON 445 Four-Wheel-Tractor, the Actual Field Capacity was 1.639 ha/h or 0.61 h/ha at a Travelling Speed of 9.46 kph at tractor's gear setting of 2nd Gear – High.

Field Efficiency

Based on the Theoretical Field Capacity of 1.824 ha/h, the Field Efficiency of the tractor using the trailing disc harrow was 89.9%.

FIELD PERFORMANCE TEST

Implement Tested	:	AMPC Trailing Harrow Model: 18 x 24 hitched to MASSEY FERGUSSON Four-Wheel Tractor Model: 445
Location of Test	:	Brgy. Murong Cauayan City, Isabela
Date of Test	:	March 14, 2013

Table 2. Results of field performance test

ITEMS				TEST DATA
I.	Test conditions			
	1.	Area (m^2)	:	1970
	2.	Soil Moisture Content (%), w.b.	:	25.7
	3.	Type of Soil	:	Sandy loam
	4.	Last Crop Planted	:	Corn
	5.	Days After Harvest	:	12
II.	Field Performance			
	1.	Type of Operation	:	Plowing
	2.	Tractor's Gear Setting	:	2^{nd} Gear – High

Table 2. Results of field performance test (cont'n.)

ITEMS			TEST DATA
3.	Traveling Speed (kph)	:	9.46
4.	Ave. Depth of Tillage (mm)	:	135
5.	Ave. Width of Tillage (mm)	:	2325
6.	Actual Field Capacity		
	a) ha/h	:	1.639
	b) h/ha	:	0.61
7.	Theoretical Field Capacity		
	a) ha/h	:	1.824
	b) h/ha	:	0.55
8.	Field Efficiency (%)	:	89.9
9.	Fuel Consumption		
	a) L/h	:	24.0
	b) L/ha	:	15.4
10.	Pattern of Operation	:	Overlapping Alternation Pattern
11.	Percent Unplowed	:	16.7

OBSERVATIONS

1. Complete cutting of the furrow slice was observed when the harrow was used as a primary tillage implement (**Figure 7**).

2. The tractor was stable, i.e. it maintained a straight path, during operation (**Figure 8**).

3. Grease fittings were provided at the bearing housings.

4. A tool box (**Figure 9**) was installed on the harrow.

Figure 7. Quality of work

Figure 8. Straightness of operation

Figure 9. Tool box

N.B.

Test Engineer:

DARWIN C. ARANGUREN

Checked by:

ROMULO E. EUSEBIO

Approved for release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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